Objectives

A) Getting familiar with Synchronization palette and concept of Queue operations.

B) To simulate an example of sound recording, transmission, playback and processing.

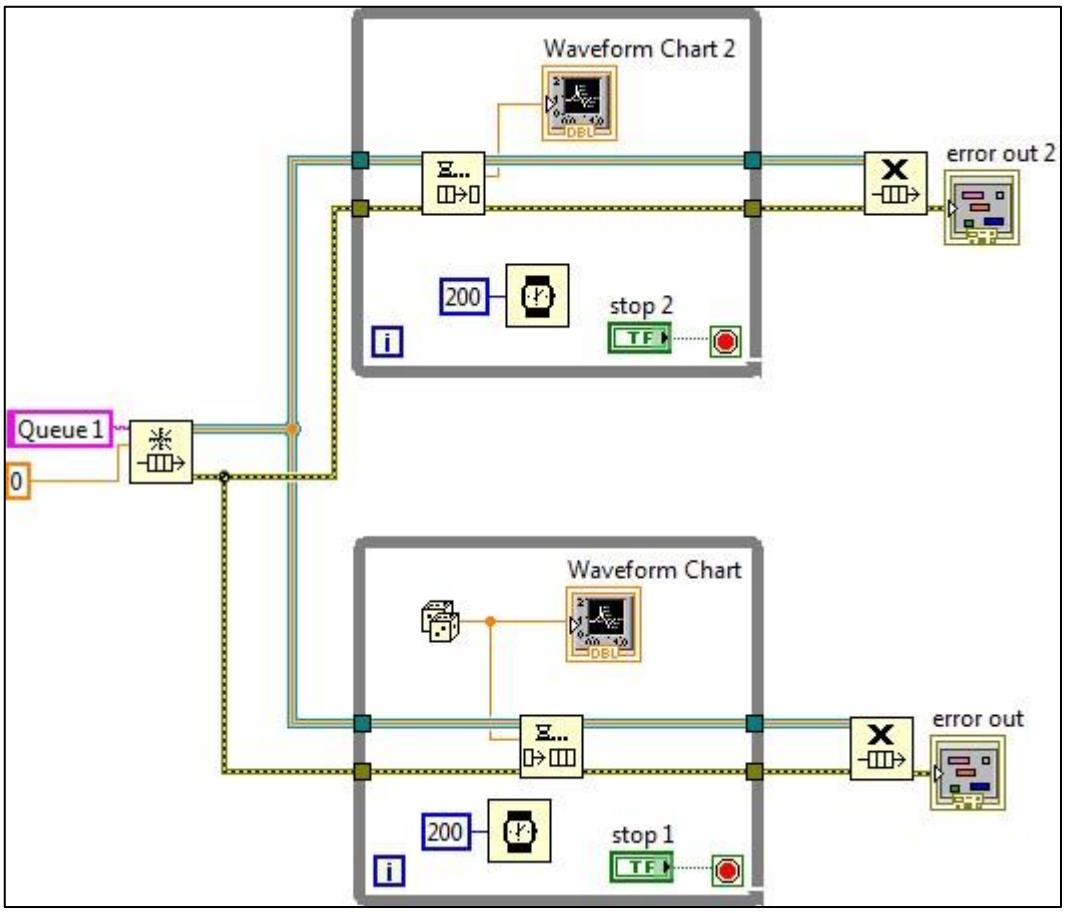
In this part of the lab, in order to have a simultaneous operation of recording and playback we use synchronization and Queue operations palettes. Synchronization is fundamental in sound transmission which is a real time task. In real time operations one should be careful about data lost. As an example, assume that one person wants to record his/her sound and speak continuously. But each function of your VI code needs time to process and it is possible that during sometimes the audio information is lost. In order to prevent losing of data “Queues” are normally used. Queues, as the name suggests, queue data. This data is stored in a buffer in memory until it is read from the queue. A queue maintains a first in/first out order of data items. For instance, the customers waiting in line at a fast food restaurant are in a queue. The first customer to arrive is going to be the first one served. The advantage of using a queue is that when you need to have to sampling rate. Items can be previewed (read) from the queue without being removed from the queue or "Dequeued". However, once dequeued the data item will no more be available on the queue. Adding items to a queue is called ‘Enqueuing’. The fundamental advantage of queue is that they buffer and store all information written to them until it is read so nothing is lost.

II. Prelab3 (Queuing)

In this experiment you need to make the LabVIEW block diagram and answer to its questions. To get familiar with **Queue Operations Pallete (Synchronization => Queue Operations)** of LabVIEW follow the below procedure:

1. Launch a new blank VI.
2. Add **Obtain Queue VI (Synchronization => Queue Operations)** and set the name of queue and element data type. For element data type create a constant double.

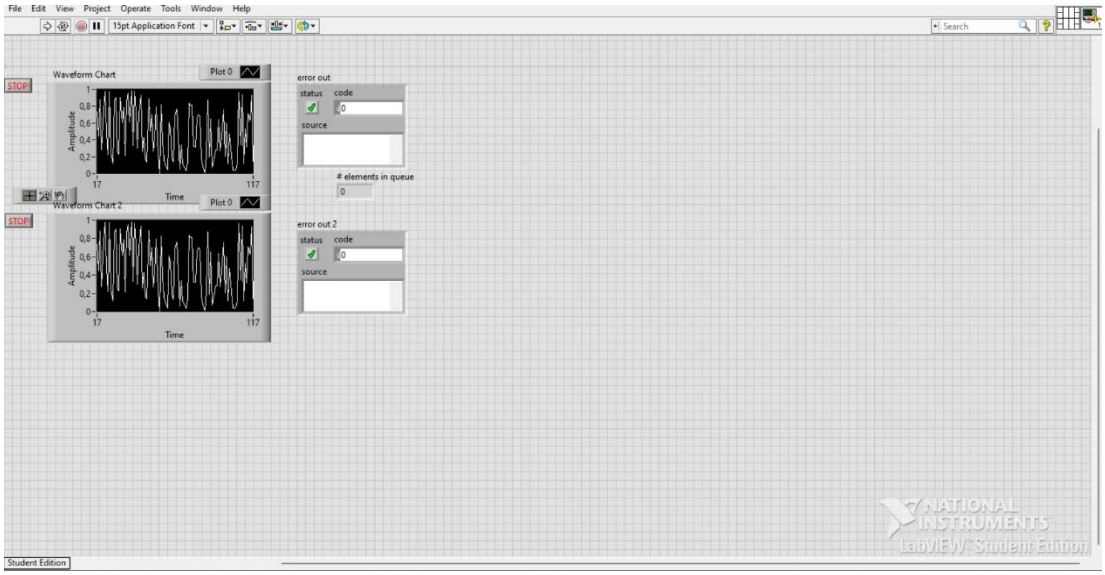
3. Create two while loop and put **Dequeue Element VI (Synchronization => Queue Operations)** in one of them and **Enqueue Element VI (Synchronization => Queue Operations)** in another one. Create the queue connections.
4. Add two **Release Queue VI (Synchronization => Queue Operations)** and put each of them outside of while loops. Create connections and set indicator for error.
5. For generating double data we use Random Number VI in numeric pallete. Put it in the while loop with the **Enqueue Element VI** and connect it to the element port of **Enqueue Element VI.** Also use a waveform chart to see the generated data.
6. At the while loop with **Dequeue Element VI**, use a waveform chart and connect it to the element port of **Dequeue Element VI.**
7. Use waits in while loops to insert delays for better observations.
8. Save the VI



Questions:

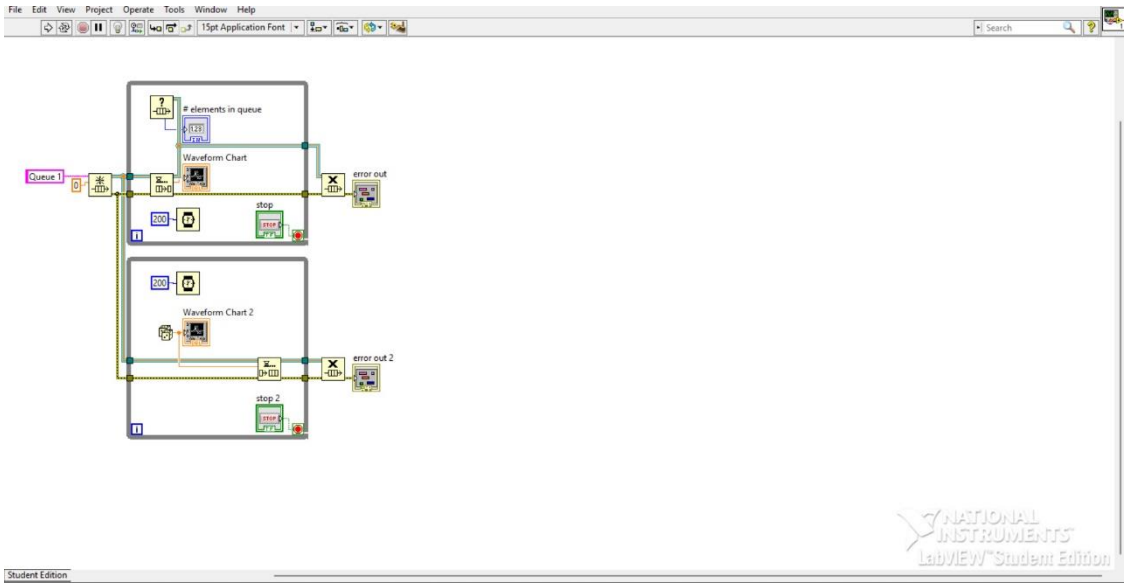
1. Run the VI and see the results. Explain how queues work and discuss about operation of Enqueue and Dequeue VIs.

The results shown on the tables are the same. The system works within the frame of FIFO, dequeue and enqueue functions enable that. Enqueuing is the function that places data on the graph and dequeue function is the one that removes data from the graph. Also, delay time (200ms) indicates the interval between two adjacent samples.



2. Insert **Get Queue Status VI (Synchronization => Queue Operations)** in one of the while loops and make the connections. Create an indicator to track number of elements in queue. How much is that?

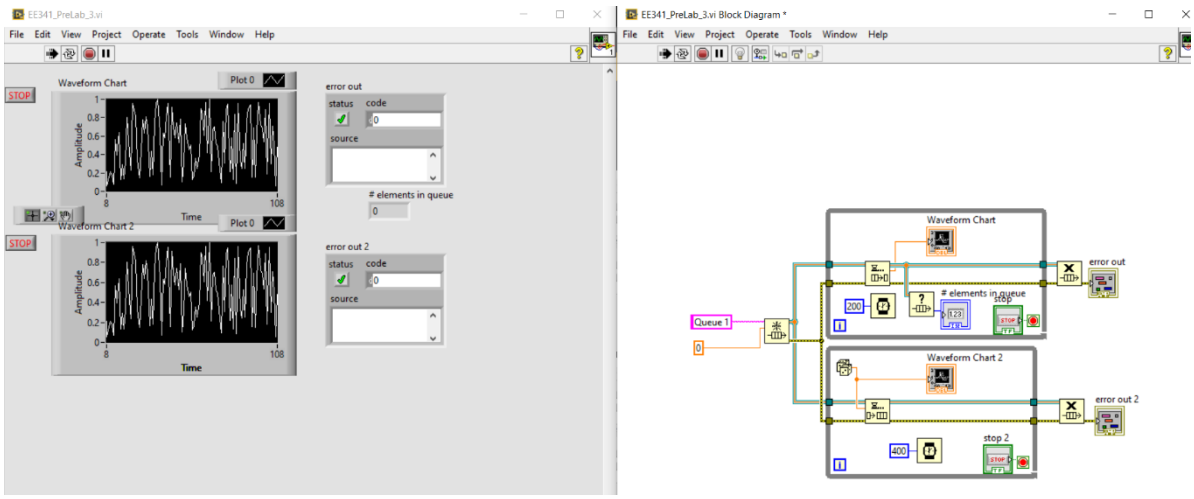
When we add the Get Queue Status, the samples can be seen but in this case the samples in the queue are constantly 0 enqueuing and dequeuing happens at the same time.





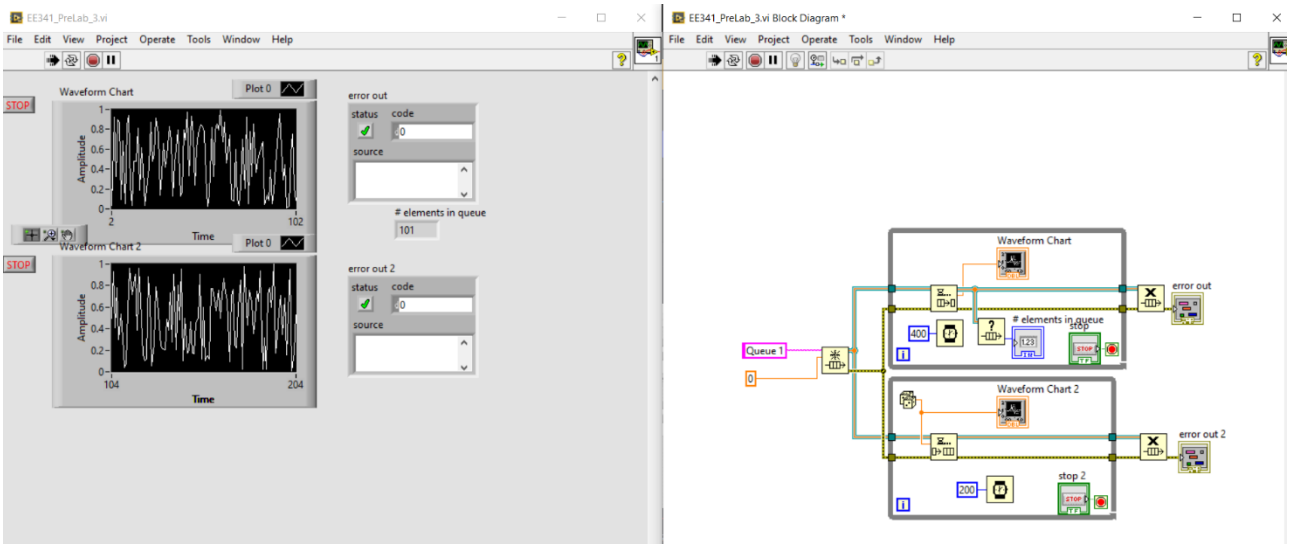
3. Increase the delay time of while loop with Dequeue VI from 200 to 400. Run the VI and observe number of elements in queue. Is it increasing? Explain the reason.

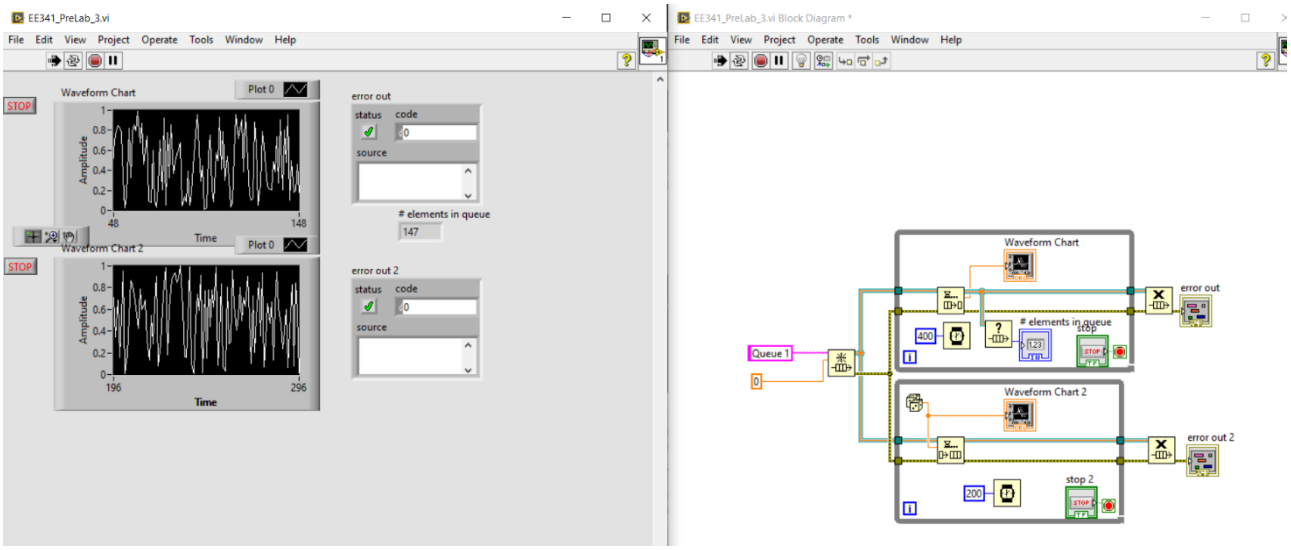
When dequeue loop interval is increased from 200 to 400 the number of samples in the queue increase at a constant rate. Speed of the dequeue plot decreases so, the graph shows half the number of samples of the enqueue graph.

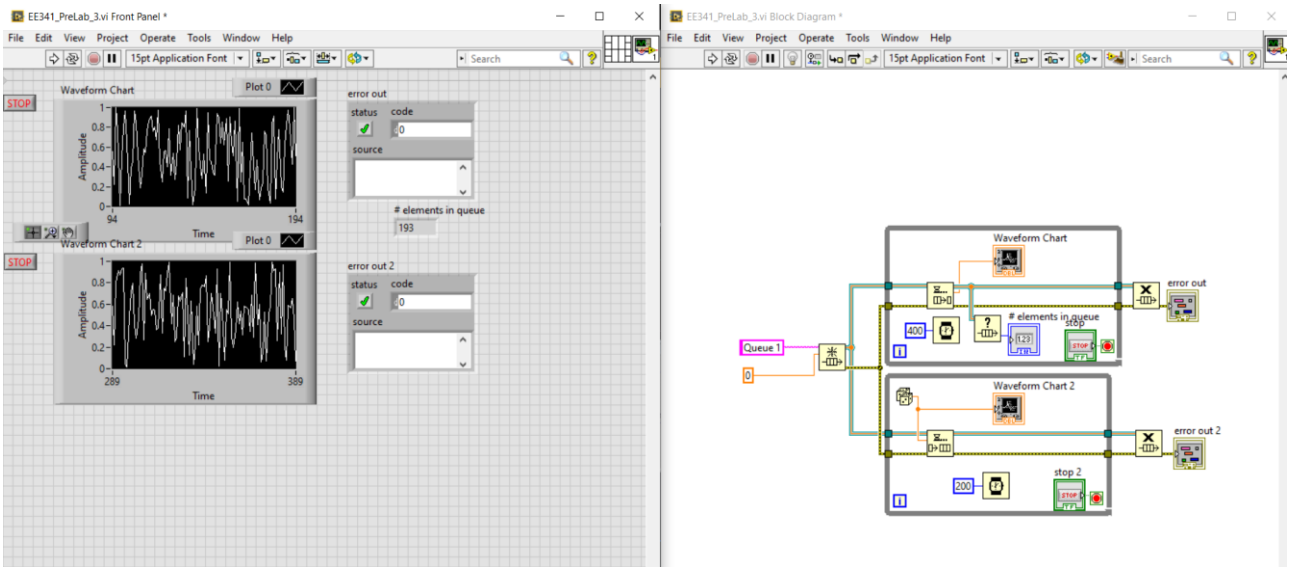


4. Reset the delays to 200. Now increase the delay time of while loop with Enqueue VI from 200 to 400. Explain your observations about number of elements in the queue.

The enqueue speed is higher than the dequeue speed so, the number of elements in queue gradually increases as time goes on.


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Queue 1
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error out 2
stop 2
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